

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `c072dd74-d017-4632-8d06-d46a8a79b899`
- **Timestamp:** 2026-09-05T15:54:48.588207Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** Which statutory provision did the Court construe in [2023] 1 S.C.R. 1169, and with what effect?
- **Answer:** I do not have enough information to answer this.
- **Health Score:** 0.07
- **Primary Issue:** RETRIEVAL_MISS
- **Claims Supported:** 0 / 1
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
---	---	---	---
CORPUS	UNKNOWN	0.5	Corpus relevance is inconclusive.
RETRIEVER	FAIL	0.85	No retrieved chunks provide support, and retrieval scores strongly indicate a lack of useful evidence.
GENERATOR	UNKNOWN	0.5	1 generated claims are unsupported, but retrieval quality is ambiguous.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 0/6 chunks used (0%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 1 (Supported: 0, Partial: 0, Unsupported: 1, Contradicted: 0)
- **Grounding Score:** 0.00
- **Evidence Coverage:** 0.17
- **Average Entailment:** 0.01
- **Latencies:** Retrieval (2347.0ms) | Generation (8420.6ms) | Verification (3419.3ms)

5. RAGAS Metrics

- **Faithfulness:** 0.0000
- **Answer Relevancy:** 0.3335
- **Context Precision:** 0.0000
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[63556e33-77a0-43d3-ac6f-4288092ffc01]** (UNSUPPORTED): I do not have enough information to answer this.
- *Evidence (Chunk 99b5577e-dfb6-4a90-af68-25598f25a30b):* 205.

7. Root Cause Analysis

- **Primary Cause:** RETRIEVAL_MISS
- **Confidence:** 0.88
- **Reasoning:**
 - Starting root cause analysis for trace c072dd74-d017-4632-8d06-d46a8a79b899.
 - Skipping CORPUS: status is UNKNOWN (insufficient evidence).
 - Failure at RETRIEVER: RETRIEVAL_MISS (confidence=0.85). Observation: No retrieved chunks provide support, and retrieval scores strongly indicate a lack of useful evidence.
 - Skipping CHUNKING: no observable state in matrix.
 - Skipping GENERATOR: status is UNKNOWN (insufficient evidence).
 - GROUNDING passed (confidence=0.90).
 - Primary root cause: RETRIEVAL_MISS at RETRIEVER (earliest upstream failure, confidence=0.85).
 - Analysis complete. Primary cause: RETRIEVAL_MISS.

8. Corrective Action Plan

- **[IMMEDIATE] Increase Retrieval Top-K**
 - Description: Increase the number of retrieved documents before providing context to the LLM.
 - Expected Improvement: Higher recall; missing context is caught within the expanded window. (Metric: Verification support)
- **[SHORT_TERM] Implement Hybrid Search (Dense + Sparse)**
 - Description: Add BM25 lexical search and combine with semantic vector search using Reciprocal Rank Fusion.
 - Expected Improvement: Near-perfect recall for exact-match terminology. (Metric: Retrieval hit rate for exact entity queries)
- **[EXPERIMENTAL] Deploy Cross-Encoder Reranker**
 - Description: Pass the Top-N results through a Cross-Encoder to re-score query-document relevance.
 - Expected Improvement: Drastically improved precision at Top-1 and Top-3. (Metric: Precision@3 increases by at least 20%)
- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
 - Description: Retrieval efficiency is low (0%); consider reducing top-k or tightening the reranker score cutoff.
 - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** RETRIEVAL_MISS
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** RETRIEVAL_MISS
- **Supporting Claims:** 63556e33-77a0-43d3-ac6f-4288092ffc01
- **Supporting Chunks:** 99b5577e-dfb6-4a90-af68-25598f25a30b

10. Overall Assessment

- **Major Strength:** GROUNDING is healthy (confidence=90%): No claims are contradicted by retrieved evidence.
- **Major Weakness:** RETRIEVAL_MISS
- **Next Priority:** Increase Retrieval Top-K
- **Recommendation:** Pipeline requires architectural adjustments across multiple stages.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/c072dd74-d017-4632-8d06-d46a8a79b899.json

✓ ClaimSet

artifacts/claim_sets/c072dd74-d017-4632-8d06-d46a8a79b899.json

✓ VerificationResults

artifacts/verification/c072dd74-d017-4632-8d06-d46a8a79b899.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/c072dd74-d017-4632-8d06-d46a8a79b899.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/c072dd74-d017-4632-8d06-d46a8a79b899.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/c072dd74-d017-4632-8d06-d46a8a79b899.json

✓ DiagnosticEvaluationReport

artifacts/reports/c072dd74-d017-4632-8d06-d46a8a79b899.json